

Proteins with multiple G protein-coupled receptor domains

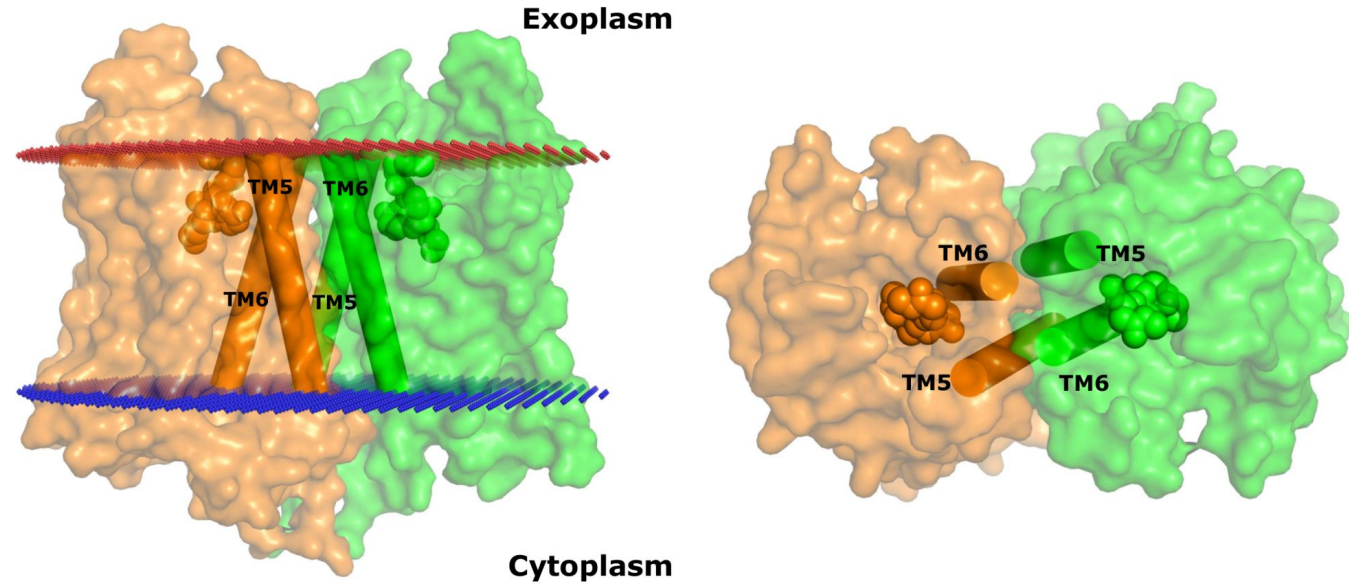
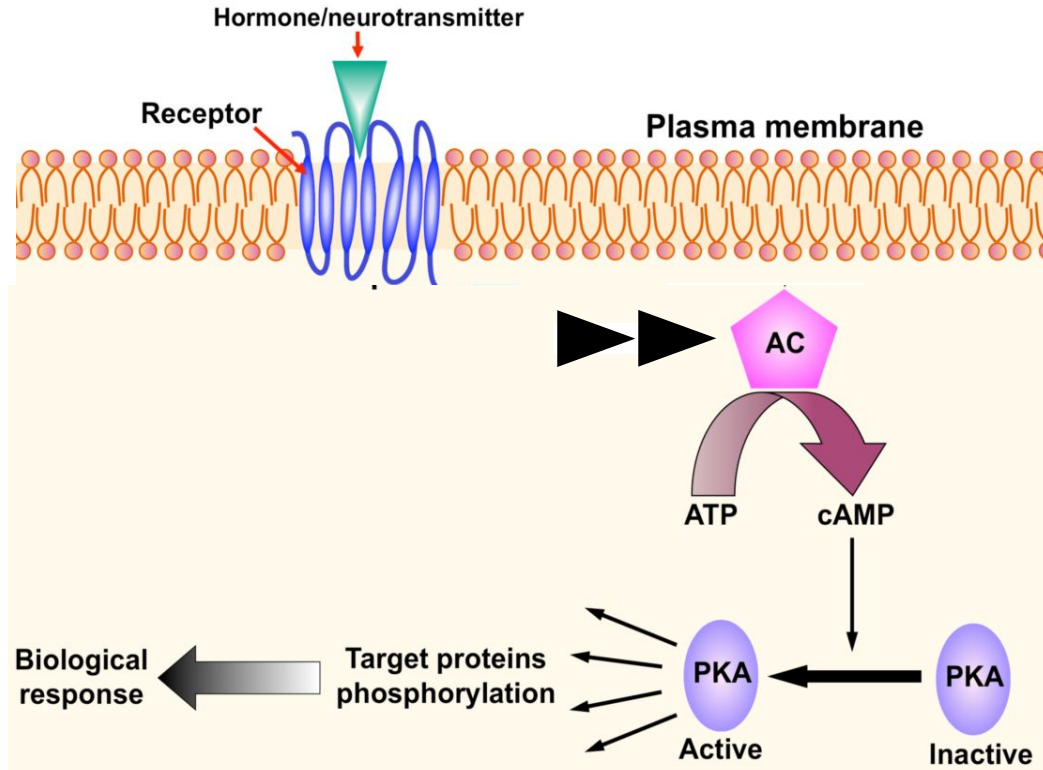
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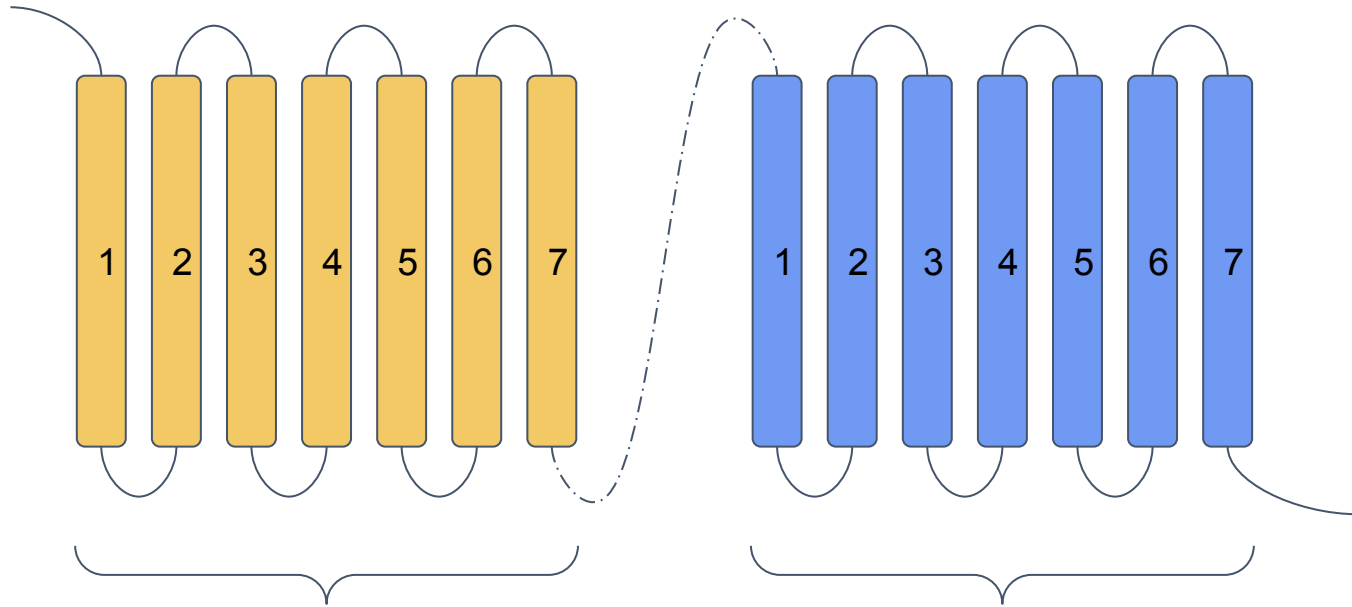
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GPCR 101

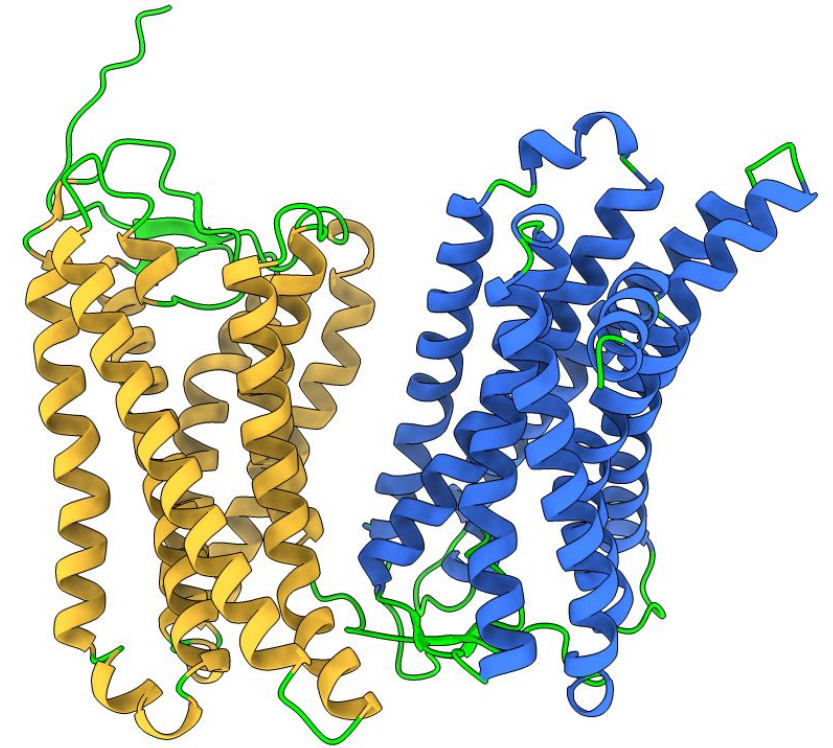


μ -opioid receptor (6TM* GPCR) in its dimer form. Parallel red and blue planes depict membrane placement, the spherical molecules are the ligands.

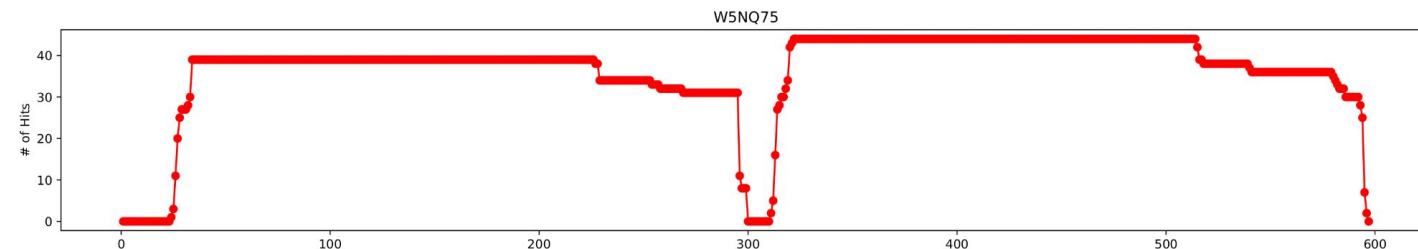
Finding proteins with multi-GPCR domains



separate ECOD GPCR domain match



W5NQ75, a possible multi-domain GPCR from sheep (*Ovis Aries*)



Tree of life distribution of homologues of the 57 multi-GPCR proteins



So far, multi-GPCR domains are only homologous to muscarinic acetylcholine receptor M3, adenosine receptor, rhodopsin, and orexin receptor type 2/ chemokine receptor type 4.

By homology, most are olfactory receptors, and some are chemokine receptors

Single-domain GPCRs are broadly distributed in eukaryotes: primitive yeast and choanoflagellate □ animals, **but multi-GPCRs only in Chordata species**

Summary and outlook

- Many putative proteins with multi-GPCR domains
- From fish to mammals up to monkeys
- Annotated mostly as olfactory receptors; few chemokine receptors
- None of them prevail, so probably not indispensable
- What is their unique contribution?
- Mechanisms?
- Multi-barrel and multi-GPCR domains... What other multi-domain membrane proteins are there?